LINEAR PROGRAMMING MAXIMIZATION STUDY GUIDE

Worked Examples • Graphical Solutions • Practice

Learning goal
Build a two-variable model, draw the feasible region, calculate every corner, choose the greatest feasible profit, and explain the result in business terms.

What linear programming does

Linear programming (LP) chooses the best use of limited resources. In a maximization problem, we seek the largest possible profit while satisfying every resource and demand limit. This guide works through the three examples in 2.2 Max Q&A.pdf.

Think of it this way
Decision variables tell us what to choose. Constraints tell us what is allowed. The objective tells us which allowed choice is best.

Key terms you must know

Term	Meaning	Example
Decision variable	An unknown quantity we choose.	x_1 = chairs per day
Objective function	The quantity we maximize.	$Z = 400x_1 + 100x_2$
Constraint	A limit that must be satisfied.	$8x_1 + 10x_2 \leq 80$
Nonnegativity	Production cannot be negative.	$x_1 \geq 0$ and $x_2 \geq 0$
Feasible region	All points satisfying every constraint.	The shaded overlap
Corner point	A vertex on the feasible boundary.	Intersection of two boundaries
Binding constraint	A limit used exactly at a solution.	Resource used = available
Slack	Unused resource or demand allowance.	Available minus used

How to use this guide

Study the method first. For each example, cover the corner table and try the calculation yourself. Then use the interpretation, practice, quiz, and answer key to check your understanding.

Source note: Original inputs and LP models are preserved. New graphs and calculations were checked independently. Pinewood's exact resource intersection is used before rounding. Whole-unit production is discussed separately.

A reliable graphical method

Step 1 Define the variables

State exactly what x_1 and x_2 represent, including units and the time period.

Step 2 Write the objective

Multiply each unit profit by its production quantity, then add. Write “Maximize Z”.

Step 3 Write every constraint

For each resource, add the quantities used and require the total to be no greater than the available amount. Include demand limits and nonnegativity.

Step 4 Draw the boundary lines

Temporarily replace ≤ with =. Find two points on each boundary and join them. For $x_1 = k$ draw a vertical line; for $x_2 = k$ draw a horizontal line.

Step 5 Shade the feasible overlap

Test a point off the boundary, often (0,0). Shade the side that satisfies the inequality. Keep only the overlap of every allowed side in the first quadrant.

Step 6 Find all feasible corners

Use exact algebra for intersections. Reject any intersection that fails even one constraint. Axis intercepts are not automatically feasible.

Step 7 Compare and interpret

Evaluate Z at every feasible corner. Choose the largest value, substitute the chosen quantities into every constraint, and explain profit and unused resources.

Why corner points work

For these nonempty bounded polygon regions, a linear objective reaches its maximum at a corner. If two adjacent corners tie for the greatest value, every point along their connecting edge is also optimal.

A small intercept demonstration

$$2x_1 + 3x_2 = 12$$

Set $x_1 = 0$: $3x_2 = 12$, so $x_2 = 4$. Set $x_2 = 0$: $2x_1 = 12$, so $x_1 = 6$. Draw the line through (0,4) and (6,0). For $2x_1 + 3x_2 \le 12$, the origin satisfies $0 \le 12$, so shade its side.

Important condition

The graphical method below treats quantities as continuous: fractions are allowed. If items must be whole, add integer restrictions and solve that revised problem. Rounding an LP answer is not a general solution method.

Worked example 1 Assembly line production

A company makes two products on two assembly lines. Line 1 has 100 available hours and line 2 has 42. Each product uses 10 hours on line 1. On line 2, product 1 uses 7 hours and product 2 uses 3. Profit is $6 per unit of product 1 and $4 per unit of product 2.

A Organize the data and build the model

Resource or profit	Product 1	Product 2	Available
Line 1 hours per unit	10	10	100 hours
Line 2 hours per unit	7	3	42 hours
Profit per unit	$6	$4	Maximize

Let x_1 = units of product 1 and x_2 = units of product 2 in the period covered by the available hours.

$$\textbf{Maximize } Z = 6x_1 + 4x_2$$

$$10x_1 + 10x_2 \le 100 \quad \text{and} \quad 7x_1 + 3x_2 \le 42$$

$$x_1 \ge 0 \quad \text{and} \quad x_2 \ge 0$$

For example, $7x_1$ measures product 1's use of line 2 in hours. Adding $3x_2$ gives total line 2 use, which cannot exceed 42 hours.

B Find intercepts and choose the allowed sides

Boundary	At $x_1 = 0$	At $x_2 = 0$
$10x_1 + 10x_2 = 100$	(0,10)	(10,0)
$7x_1 + 3x_2 = 42$	(0,14)	(6,0)

Both inequalities accept the origin, so shade below both lines in the first quadrant. On the horizontal axis the tighter limit is $x_1 \le 6$; on the vertical axis it is $x_2 \le 10$.

C Calculate the intersection

Divide the line 1 equation by 10: $x_1 + x_2 = 10$. Hence $x_2 = 10 - x_1$.

$$7x_1 + 3(10 - x_1) = 42$$

$$7x_1 + 30 - 3x_1 = 42 \rightarrow 4x_1 = 12 \rightarrow x_1 = 3$$

$$x_2 = 10 - 3 = 7$$

Before you turn the page
Can (10,0) be a feasible corner? Check its line 2 use. Can (0,14) be feasible? Check its line 1 use.

Example 1 Graph and optimal solution

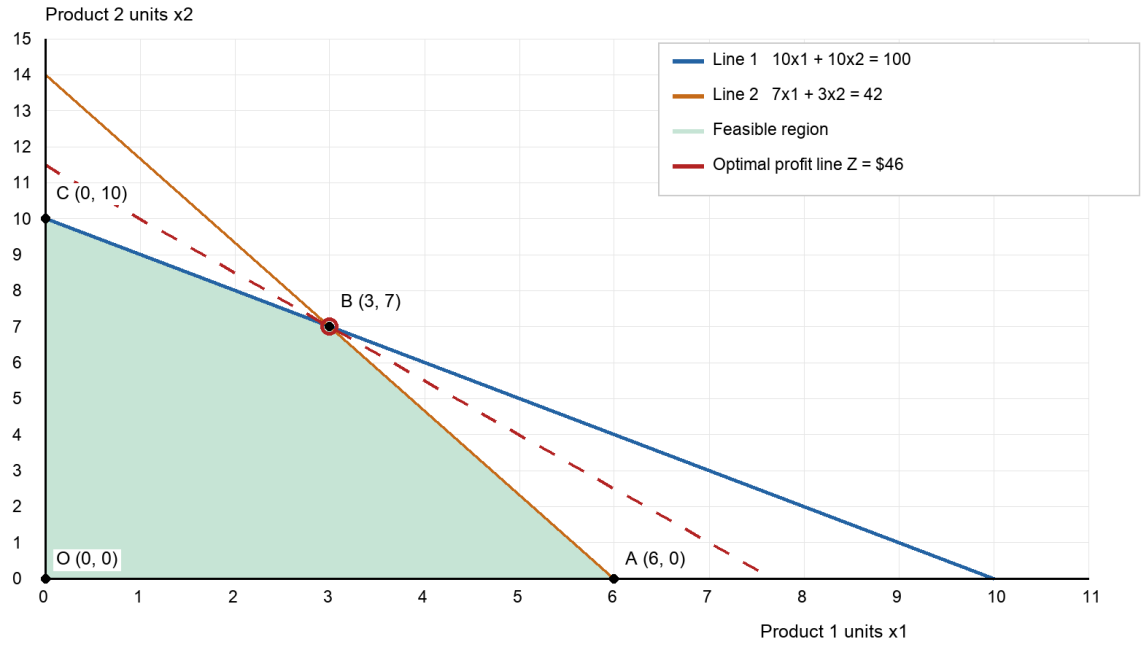


Figure 1. The green polygon is the shared feasible region. O, A, B, and C are its corners. The dashed profit line touches the region at the optimal corner B.

D Evaluate every feasible corner

Corner	Coordinates	Profit calculation	Z
O	(0,0)	6(0) + 4(0)	$0
A	(6,0)	6(6) + 4(0)	$36
B	(3,7)	6(3) + 4(7)	$46
C	(0,10)	6(0) + 4(10)	$40

Answer and interpretation

Produce 3 units of product 1 and 7 units of product 2. The maximum profit is $46 for the stated resource period.

E Check feasibility and resource use

Line 1: 10(3) + 10(7) = 100 hours; slack = 100 − 100 = 0.
Line 2: 7(3) + 3(7) = 42 hours; slack = 42 − 42 = 0.
Both quantities are nonnegative. Both resource constraints bind.

Beginner lesson

The higher-profit product is not automatically the only product to make. Product 1 uses more of scarce line 2 time; a mixed production plan gives more total profit.

Worked example 2 Pinewood Furniture Company

Pinewood produces chairs and tables each day using 80 hours of labor and 36 pounds of wood. Chair demand is at most 6 per day. A chair requires 8 labor hours and 2 pounds of wood; a table requires 10 labor hours and 6 pounds of wood. Profit is $400 per chair and $100 per table.

A Organize the data and build the model

Resource or profit	Chair	Table	Daily limit
Labor hours per unit	8	10	80 hours
Wood pounds per unit	2	6	36 pounds
Profit per unit	$400	$100	Maximize
Demand	At most 6	No stated cap	6 chairs

Let x_1 = chairs per day and x_2 = tables per day.

$$\textbf{Maximize } Z = 400x_1 + 100x_2$$

$$8x_1 + 10x_2 \le 80 \quad \text{and} \quad 2x_1 + 6x_2 \le 36$$

$$x_1 \le 6 \quad \text{with} \quad x_1 \ge 0 \text{ and } x_2 \ge 0$$

B Draw the boundaries

Boundary	Vertical axis intercept	Horizontal axis intercept
$8x_1 + 10x_2 = 80$	(0,8)	(10,0)
$2x_1 + 6x_2 = 36$	(0,6)	(18,0)
$x_1 = 6$	Vertical line	Through (6,0)

Shade below both resource lines, to the left of $x_1 = 6$, and in the first quadrant. The feasible axis endpoints are (6,0) and (0,6).

C Calculate the two nonaxis corners

Chair cap with labor: $8(6) + 10x_2 = 80 \rightarrow 10x_2 = 32 \rightarrow x_2 = 3.2$. Check wood: $2(6) + 6(3.2) = 31.2 \le 36$. This gives B = (6,3.2).

Labor with wood: multiply $2x_1 + 6x_2 = 36$ by 4, giving $8x_1 + 24x_2 = 144$. Subtract $8x_1 + 10x_2 = 80$: $14x_2 = 64$, so $x_2 = 32/7$. Substitute into wood: $2x_1 = 36 - 192/7 = 60/7$, so $x_1 = 30/7$.

$$C = (30/7, 32/7) \approx (4.2857, 4.5714)$$

The cap with wood gives (6,4), but labor use is 88 > 80, so reject this intersection.

Example 2 Graph and optimal solution

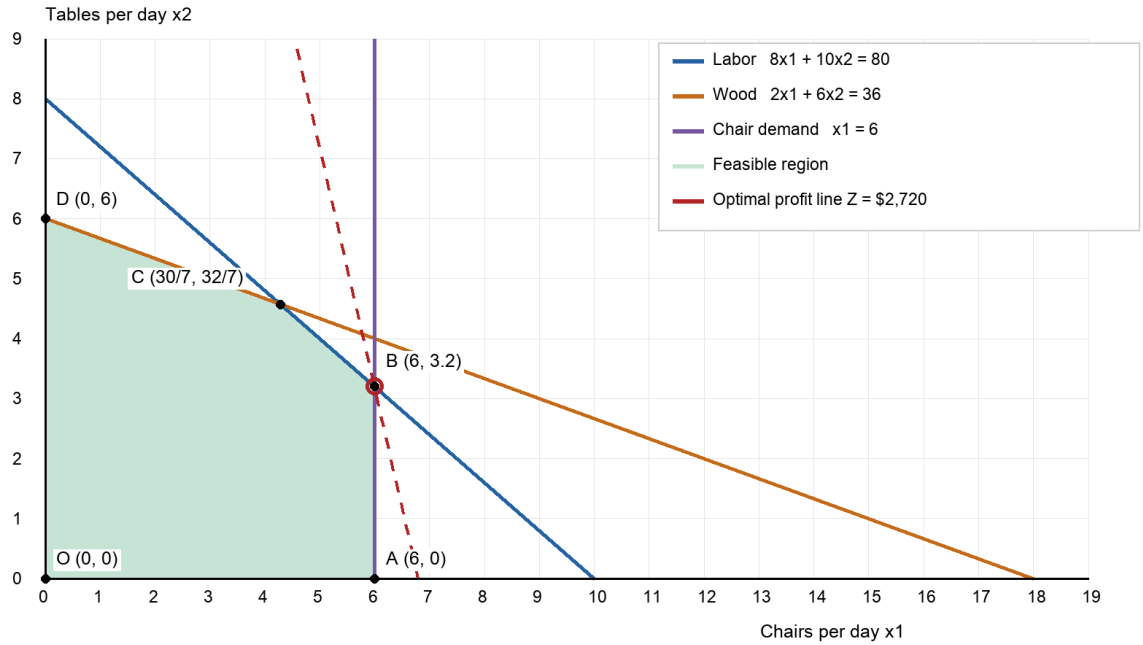


Figure 2. The wood line's intercept at (18,0) is shown for plotting; it is outside the feasible region. The chair cap makes the right edge vertical.

D Evaluate every feasible corner

Corner	Coordinates	Profit calculation	Z
O	(0,0)	400(0) + 100(0)	$0
A	(6,0)	400(6) + 100(0)	$2,400
B	(6,3.2)	400(6) + 100(3.2)	$2,720
C	(30/7,32/7)	(12,000 + 3,200)/7	$2,171.43
D	(0,6)	400(0) + 100(6)	$600

Answer and interpretation

In the continuous LP model, produce 6 chairs and 3.2 tables per day for a maximum daily profit of $2,720.

Labor: 8(6) + 10(3.2) = 80 hours, slack 0. Wood: 2(6) + 6(3.2) = 31.2 pounds, slack 4.8 pounds. Chair demand: 6 − 6 = 0 unused allowance. Labor and chair demand bind; wood does not.

Whole furniture units

If daily production must be whole units, the best plan is 6 chairs and 3 tables, earning $2,700. With at most 5 chairs, even the loose bound $x_2 \le 6$ gives profit ≤ $2,600. With 6 chairs, labor permits at most 3 whole tables. This proves the whole-unit optimum.

Rounding note: the source prints C as (4.29,4.57), but calculates profit from the exact intersection. Substituting those rounded coordinates gives $2,173, so retain fractions until the final profit is rounded.

Common mistakes and formula cheat sheet

Mistake	How to fix it
Using revenue in place of profit	Use the stated profit per unit in the objective.
Forgetting a demand cap	Write $x_1 \le 6$ for chairs and $x_2 \le 4$ for bracelets.
Drawing only one allowed region	Keep the overlap satisfying every inequality.
Treating all intercepts as corners	Check each candidate against every constraint.
Assuming every intersection is feasible	Reject (6,4) in Pinewood because labor use is 88.
Reading precise answers from the graph	Use algebra to calculate intersections.
Rounding too early	Keep fractions until the final displayed result.
Rounding a fractional LP solution blindly	If whole units are required, verify an integer optimum.
Giving profit without production quantities	State x_1, x_2, units, profit, and the time period.

Quick formula cheat sheet

Task	Formula or rule
Profit to maximize	$Z = p_1x_1 + p_2x_2$
Resource limit	$a_1x_1 + a_2x_2 \le b$
Plot a resource boundary	$a_1x_1 + a_2x_2 = b$
Horizontal axis intercept	$(b/a_1, 0)$, when $a_1 \neq 0$
Vertical axis intercept	$(0, b/a_2)$, when $a_2 \neq 0$
Resource slack	$b - (a_1x_1 + a_2x_2)$
Corner comparison	Zmax = largest Z among feasible corners
Vertical and horizontal caps	$x_1 = k$ is vertical; $x_2 = k$ is horizontal

Here p_1 and p_2 are unit profits; a_1 and a_2 are resource use per unit; b is available resource. Use the original resource units when reporting slack.

Final study tip

Use the sequence: variables → objective → constraints → boundary lines → feasible overlap → corners → profits → interpretation. Check the chosen plan in the original constraints.

Practice problems

Try these without looking at the worked examples. Use the same graphical method and show substitutions. Problems 1 to 3 reuse the original models; problem 4 is a new exercise.

Practice 1 Assembly line feasibility

For the assembly model, test (4,6) against both resource limits. Find its profit. If it is infeasible, identify the violated limit and the excess resource use. Then state the best feasible corner and profit.

Practice 2 Pinewood resource checks

At the continuous optimum (6,3.2), calculate labor slack, wood slack, and unused chair demand allowance. Explain why (6,4) is infeasible. If whole furniture items are required, state the optimal daily plan and profit.

Practice 3 Jewelry corner comparison

Compare (2,4) and (4,3): calculate the profit and resource use of each. Which gives higher profit? At the optimal plan, is bracelet demand binding?

Practice 4 A new small production model

A workshop makes products A and B. Each A earns \$5 and each B earns \$4. Resource limits give $x_1 + x_2 \leq 8$ and $2x_1 + x_2 \leq 10$, where x_1 = units of A and x_2 = units of B. Quantities are nonnegative. Formulate the model, graph the feasible region, calculate all corners, and find the maximum profit.

Working space: sketch the axes, label each boundary, and list the corners before calculating profit.

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Self check quiz

1. A feasible point must satisfy: (a) only the objective (b) every constraint (c) one resource limit.

2. The line $x_1 = 6$ is: (a) horizontal (b) vertical (c) diagonal.

3. If resource use equals its availability, its slack is: (a) zero (b) negative (c) the full availability.

4. True or false: The intersection of any two boundaries is automatically feasible.

5. True or false: A fractional LP answer can always be rounded to obtain the optimal whole-unit answer.

6. Why should you use exact coordinates before calculating a corner's profit?

Answer key

Practice 1

At (4,6), line 1 use = 10(4) + 10(6) = 100 hours. Line 2 use = 7(4) + 3(6) = 46 hours, exceeding 42 by 4 hours. Profit would be $48, but this point is infeasible. The best feasible corner is (3,7), with profit $46.

Practice 2

At (6,3.2), labor slack = 80 − [8(6) + 10(3.2)] = 0 hours. Wood slack = 36 − [2(6) + 6(3.2)] = 4.8 pounds. Unused chair allowance = 6 − 6 = 0. At (6,4), labor use is 48 + 40 = 88 hours, exceeding 80 by 8. The optimal whole-unit plan is 6 chairs and 3 tables, with daily profit $2,700.

Practice 3

Plan	Gold used	Platinum used	Profit
(2,4)	14 ounces	20 ounces	$2,200
(4,3)	18 ounces	20 ounces	$2,400

Both plans satisfy the bracelet cap. The plan (4,3) earns $200 more. Bracelet demand is not binding: 4 − 3 = 1 bracelet of unused allowance remains.

Practice 4

Maximize $Z = 5x_1 + 4x_2$

$$x_1 + x_2 \le 8; \quad 2x_1 + x_2 \le 10; \quad x_1 \ge 0; \quad x_2 \ge 0$$

Plot the first boundary through (8,0) and (0,8), and the second through (5,0) and (0,10). Shade the common side containing the origin in the first quadrant. Subtract $x_1 + x_2 = 8$ from $2x_1 + x_2 = 10$: $x_1 = 2$, then $x_2 = 6$.

Corner	Profit calculation	Z
(0,0)	5(0) + 4(0)	$0
(5,0)	5(5) + 4(0)	$25
(2,6)	5(2) + 4(6)	$34
(0,8)	5(0) + 4(8)	$32

Produce 2 units of A and 6 units of B for maximum profit $34. Resource use: 2 + 6 = 8 and 2(2) + 6 = 10, so both constraints bind.

Self check answers

1 b. Feasibility requires every constraint. 2 b. A fixed x_1 value gives a vertical line. 3 a. Used exactly means zero slack. 4 False. Other limits may exclude the intersection. 5 False. Rounding can violate a limit or miss a better integer plan. 6 Rounded coordinates can distort feasibility and profit comparisons; exact fractions prevent premature rounding errors.

Optional Python corner checks

This short example uses only standard Python. It checks feasibility and compares the known corners from the hand calculations. It verifies those candidates; it does not discover missing corners or solve an arbitrary LP. You should still draw the graph and calculate all corners first.

```
from fractions import Fraction as F

models = [
    ("Assembly", [6, 4], [[10, 10], [7, 3]], [100, 42],
     [(0, 0), (6, 0), (3, 7), (0, 10)]),
    ("Pinewood", [400, 100], [[8, 10], [2, 6], [1, 0]],
     [80, 36, 6], [(0, 0), (6, 0), (6, F(16, 5)),
                   (F(30, 7), F(32, 7)), (0, 6)]),
    ("Jewelry", [300, 400], [[3, 2], [2, 4], [0, 1]],
     [18, 20, 4], [(0, 0), (6, 0), (4, 3), (2, 4), (0, 4)])
]

for name, profit, A, limits, corners in models:
    feasible = []
    for x, y in corners:
        allowed = x >= 0 and y >= 0 and all(
            a*x + b*y <= limit
            for (a, b), limit in zip(A, limits)
        )
        if allowed:
            feasible.append((x, y))
    best = max(feasible, key=lambda v:
               profit[0]*v[0] + profit[1]*v[1])
    z = profit[0]*best[0] + profit[1]*best[1]
    x, y = map(float, best)
    print(f"{name}: x1={x:g}, x2={y:g}, profit=${float(z):,.2f}")
```

Expected output

Assembly: x1=3, x2=7, profit=$46.00
Pinewood: x1=6, x2=3.2, profit=$2,720.00
Jewelry: x1=4, x2=3, profit=$2,400.00

How to read the code

A stores resource-use coefficients and limits stores the matching capacities. all checks every limit. max chooses the feasible corner with greatest profit. Fraction keeps Pinewood’s coordinates exact.